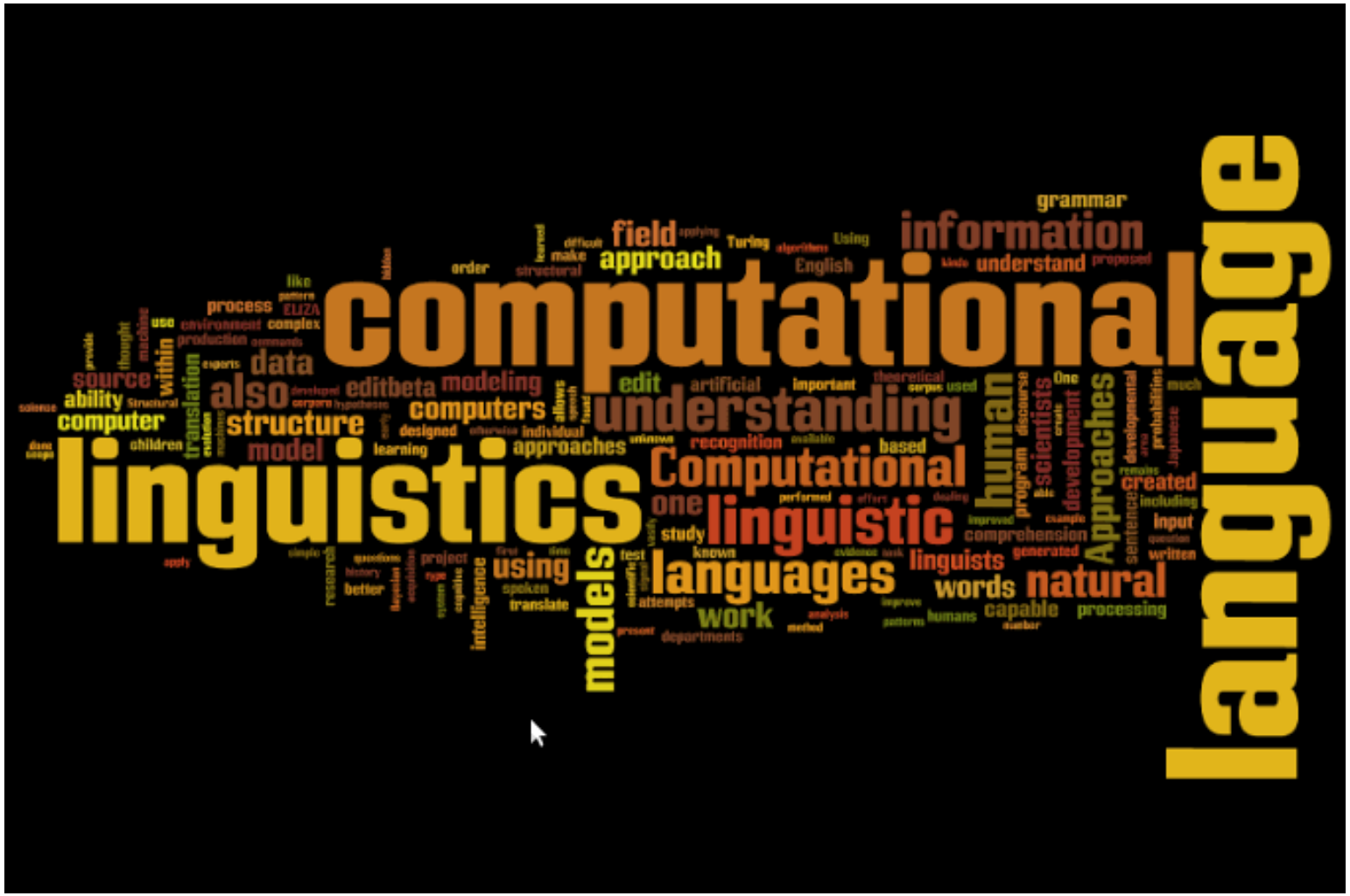
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Computational Linguistics -2

Assignment-1



Q1)

Python code to convert brown-test.txt into lower case

```
with open ("./browndata/brown-train.txt","r") as reading_file, open (“brown-test-lwr.txt”,"w") as writing_file:
    for sentences in reading_file:
        sentences=sentences.lower()
        writing_file.write(sentences)
```

Training using SRILM toolkit and Kneser Ney interpolation on (*Goodman 2001*):



Research Paper “A bit of progress in language modeling” empericaly showed that Interpolated Kneser Ney performs way better than backoff Kneser Ney

a) Trigram

Command used: ngram-count -text ./browndata/brown-train.txt -order 3 -kndiscount -interpolate -lm trigram.lm

b) Bigram

Command used: ngram-count -text ./browndata/brown-train.txt -order 2 -kndiscount -interpolate -lm bigram.lm

Python code to extract sentences from the bnc.csv file and store in a text file named ‘bnc_sentences’

```
import pandas as pd
df = pd.read_csv('bnc.csv')

# Write plain-text file for SRILM, one sentence per line, in the same order as the csv
#create a file bnc_sentences to store the required testing data sentence by sentence
with open('bnc_sentences', 'w') as f:
    for s in df['text']:
        f.write(s.strip() + '\n')
```

Python code to lower-case the letters in test file

```
with open ("srilm_clean_sentences.txt","r") as reading_doc, open ("bnc_test.txt","w")as writing_doc:
    for sentence in reading_doc:
        sentence=sentence.lower()
        writing_doc.write(sentence)
```

This step is important as 18 sentences given in the bnc.csv test file are completely in uppercase, and srilm consider all such uppercase strings as OOV’s as all the training data was in lower case. Thus, it gives 0 as the probability for all such sentences (convention) which is wrong. Lowercasing avoids these issues.





Training unigram model (to be used for the SLOG) without smoothing or interpolation technique

Command used: ngram-count -text ./browndata/brown-train-lwr.txt -order 1 -unk -addsmooth 0 -lm unigram.lm

to stop the default smoothing used by SRILM and get the required P_unigram

Obtaining the language model scores

Commands:

```
ngram -lm trigram.lm -order 3 -unk -no-eos -ppl bnc_test.txt -debug 1 >trigram_scores.txt
ngram -lm bigram.lm -order 2 -unk -no-eos -ppl bnc_test.txt -debug 1 >bigram_scores.txt
```

Note: The conclusion regarding the testing of hypothesis is mentioned after the end of Q2 (so that all sorts of data are collected).

Q2)

Python code to obtain the spearman correlation

```
import pandas as pd
from scipy.stats import spearmanr

RATINGS_FILE = "bnc.csv"
SCORES_FILE = "q1_scores.csv"


# Read ratings
ratings = pd.read_csv(RATINGS_FILE)

ratings = ratings[["text", "mean_rating"]].copy()
ratings["mean_rating"] = pd.to_numeric(
    ratings["mean_rating"],
    errors="coerce"
)

ratings["sent_idx"] = range(len(ratings))

# Read LM scores
scores = pd.read_csv(SCORES_FILE)

print("Ratings:", len(ratings))
print("Scores :", len(scores))

if len(ratings) != len(scores):
```

```
    raise ValueError(
        f"Number of sentences does not match: "
        f"ratings={len(ratings)}, scores={len(scores)}"
    )

# Match sentences
df = pd.merge(
    ratings,
    scores,
    on="sent_idx",
    how="inner"
)

score_columns = [
    "tri_logP",
    "tri_avgLogP",
    "tri_SLOR",
    "bi_logP",
    "bi_avgLogP",
    "bi_SLOR"
]

for column in score_columns:
    df[column] = pd.to_numeric(
        df[column],
        errors="coerce"
    )

df = df.dropna(
    subset=["mean_rating"] + score_columns
)
```

```
print("Valid sentences:", len(df))

# Calculate Spearman correlations
results = [ ]

for column in score_columns:
    rho, p_value = spearmanr(
        df["mean_rating"],
        df[column]
    )

    results.append({
        "Measure": column,
        "Spearman_rho": rho,
        "p_value": p_value
    })

# Display results
results_df = pd.DataFrame(results)

print("\nSpearman Correlations")
print(results_df.to_string(index=False))

# Save results
results_df.to_csv(
    "spearman_results.csv",
    index=False
)

df.to_csv(
```

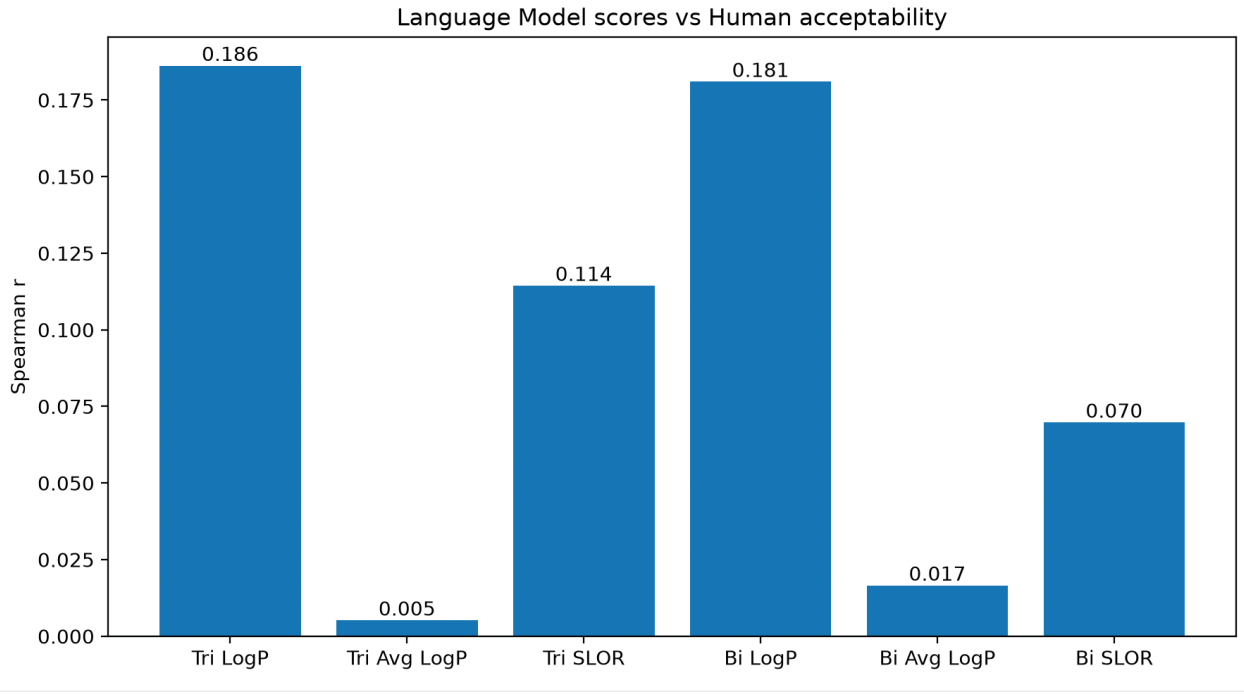
```
    "final_q1_data.csv",
    index=False
)

# Best score
best = results_df.loc[
    results_df["Spearman_rho"].idxmax()
]

print(
    f"\nBest score: {best['Measure']} "
    f"(rho = {best['Spearman_rho']:.6f})"
)
```

```
Spearman Correlations
    Measure  Spearman_rho      p_value
   tri_logP      0.186234 3.888305e-10
tri_avgLogP      0.005217 7.054735e-01
   tri_SLOR      0.114433 9.033789e-17
    bi_logP      0.181002 6.887369e-10
 bi_avgLogP      0.01660 2.007585e-01
    bi_SLOR      0.069788 4.157874e-07
```

Spearman Correlation Table (Output)



Bar graph showing visual representations

The Spearman correlation in sorted order for the models are as follows:

tri_logP > bi_logP > tri_SLOR > bi_SLOR > bi_avgLogP > tri_avgLogP

('tri' denotes a tringram, 'bi' denotes a bigram and rest symbols have their usual meanings).

Trigram.lm and tri_logP (the total log probability for trigram model) was observed to have worked the best in this experiment. This is quite contrary with the research paper where the convention devised by author of that of SLOR performed better than other models.

One reason turns out behind this is the fact that our testing was done on a test corpus, which was just 5250 sentences long, hence this corpus was not enough for SLOR to create a difference as compared to "log probability" model that worked better here. The distribution and category or words (less use of stop words like 'the', 'as', etc) and translation from different languages could also be probable reason behind the defeat of SLOR.

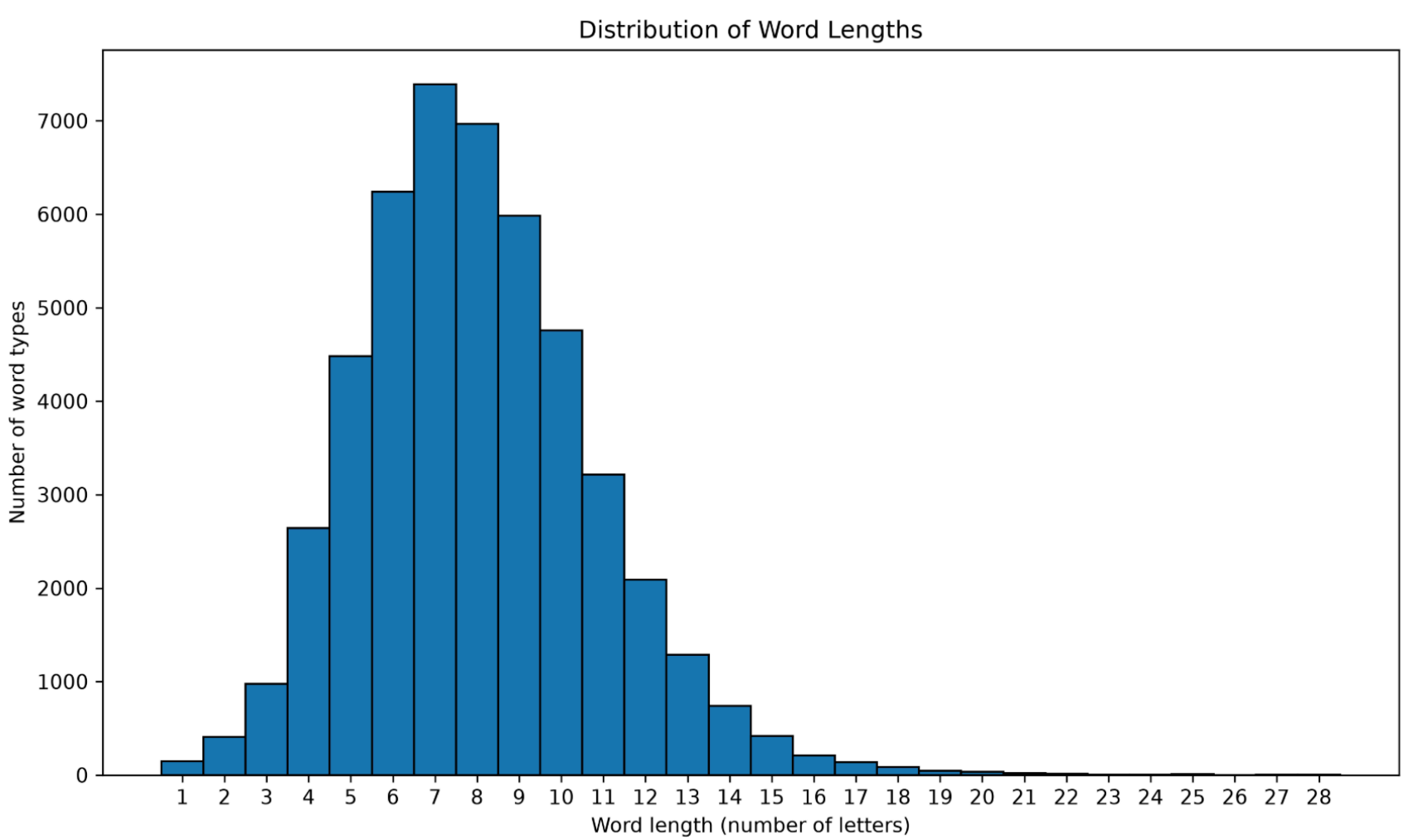
Hypothesis testing and results

The results provide weak but positive support for the hypothesis that sentence acceptability is related to language-model probability. Among the tested measures, trigram lopP performs best, but that too had a very low spearman correlation value of roughly 0.18. Probability can provide a better measure of syntactic/structural well-formedness than raw or average log-probability. However, the correlations are small, so the results do not indicate that language-model probability alone is a strong predictor of human acceptability. Human judgments are influenced by factors beyond the probability captured by these bigram and trigram models.

However, upon keen observation of the ppl scores observed (lesser perplexity higher probability) of the trigram and bigram models show that the sentences having a higher perplexity score (lower probability) was indeed having smaller mean rating of human perception, showing a small but positive correlation between the Probability and human perception of grammaticality.

Final claim: Human perception and probability are co-related and even though this co-relation was small but evident in the dataset, despite it being such a small collection of sentences. Larger datasets could be used to establish even more stronger co-relation.

Q3) The Brown corpus (train, development, and test sets) and the BNC dataset were combined into a single dataset. All words were converted to lowercase, and punctuation-only tokens were excluded. For each remaining word, word length was calculated as the **number of alphabetic letters** in the word. For example, *cat* has a length of 3, *language* has a length of 8, and *don't* has a length of 4 when only letters are counted. The frequency and length of each word were then recorded for further analysis.



Histogram showing distribution of word lengths

1) Information content of a word, $I(w) = -\log_2 P(w)$
where P(w) is the probability of the word in the combined dataset. (P(w) = f(w) / N)
Here, f(w) is the frequency of the word and N is the total number of word tokens.

Therefore, frequent words have lower information content, while rare words have higher information content. The word-level dataset contains 1,086,993 word tokens and 48,294 unique word types.

Regression Analysis

Two separate linear regression models were fitted, with word length as the dependent variable.

Model 1 used: Frequency as predictor
$L = \beta_0 + \beta_1 F$
Final output of the model : $R^2 = 0.00284$

Model 2 used : Information content as predictor
$L = \beta_0 + \beta_1 I$
Final output of the second model obtained: $R^2 = 0.05253$

The information-content model has a substantially higher R^2 value, than the frequency model (0.05253 vs 0.00284 respectively).
Thus, **information content** is a better predictor of word length than **raw frequency** for this dataset, supporting the proposed hypothesis. The information-content model explains

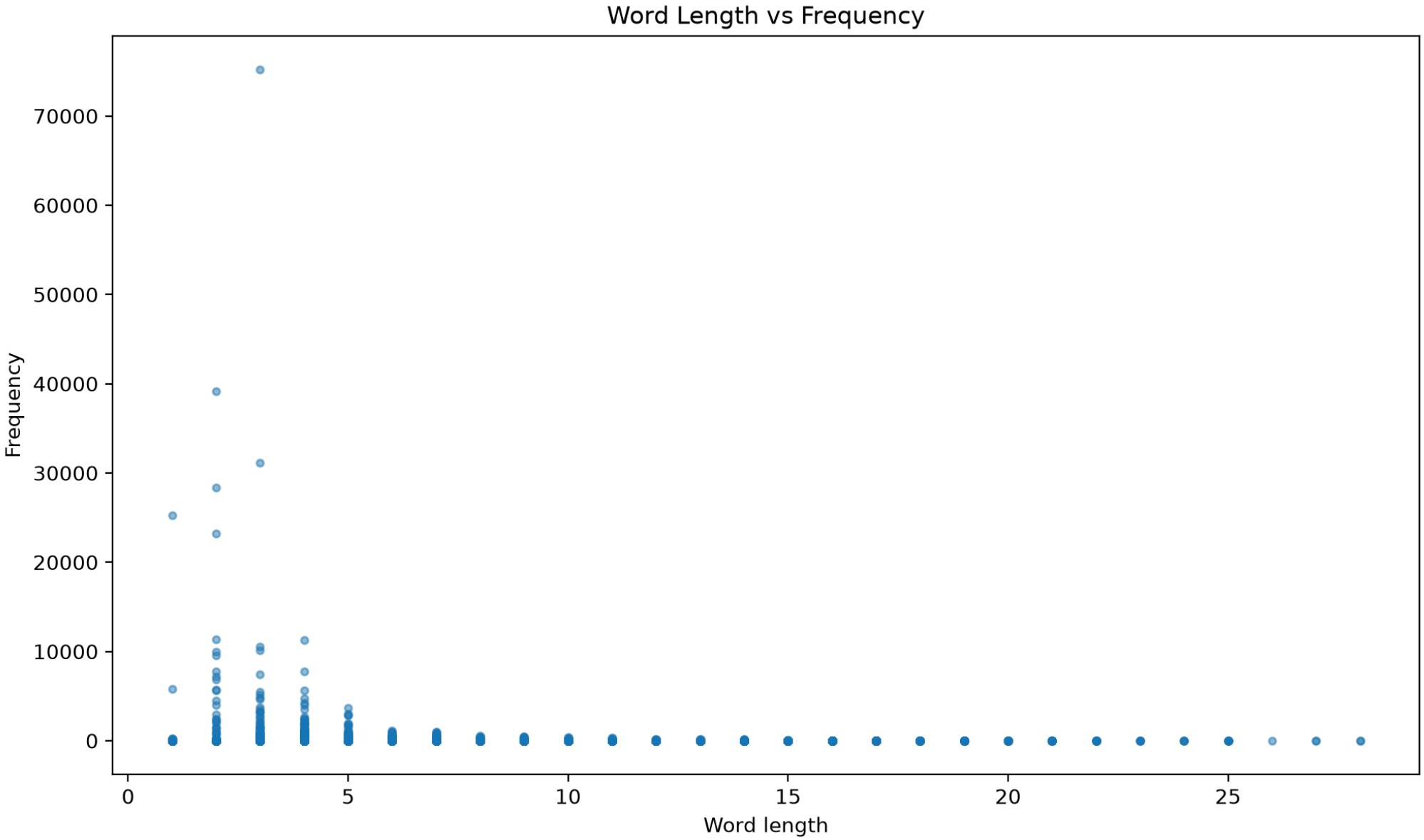
approximately 5.25% of the variation in word length, whereas the frequency model explains only approximately 0.28%.

However, the relatively low R^2 of the information-content model was relatively low (5.28%), which indicates that information content alone does not explain most of the variation in word length. It is nevertheless a considerably better predictor than raw frequency in this analysis.

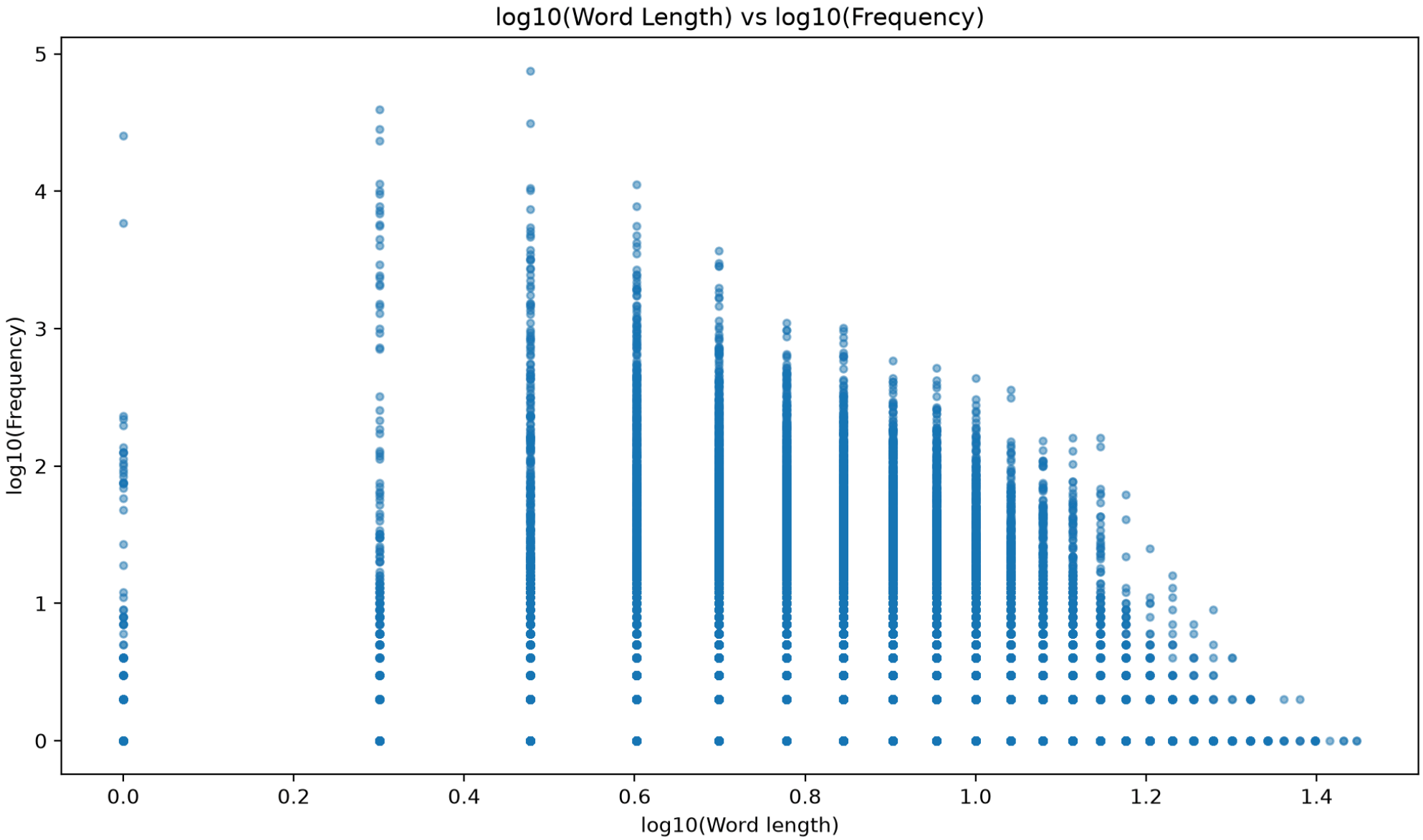
2) Single letter words are the shortest words in my dataset. These includes article “a” , and personal pronoun “I”. There frequencies are 25,272 and 5835 respectively and are common English words used in abundance throughout the dataset.

3) The Pearson correlation coefficient between word length and word frequency is $r=-0.0533$. This indicates a very weak negative correlation: more frequent words tend to be slightly shorter, but the relationship is weak.

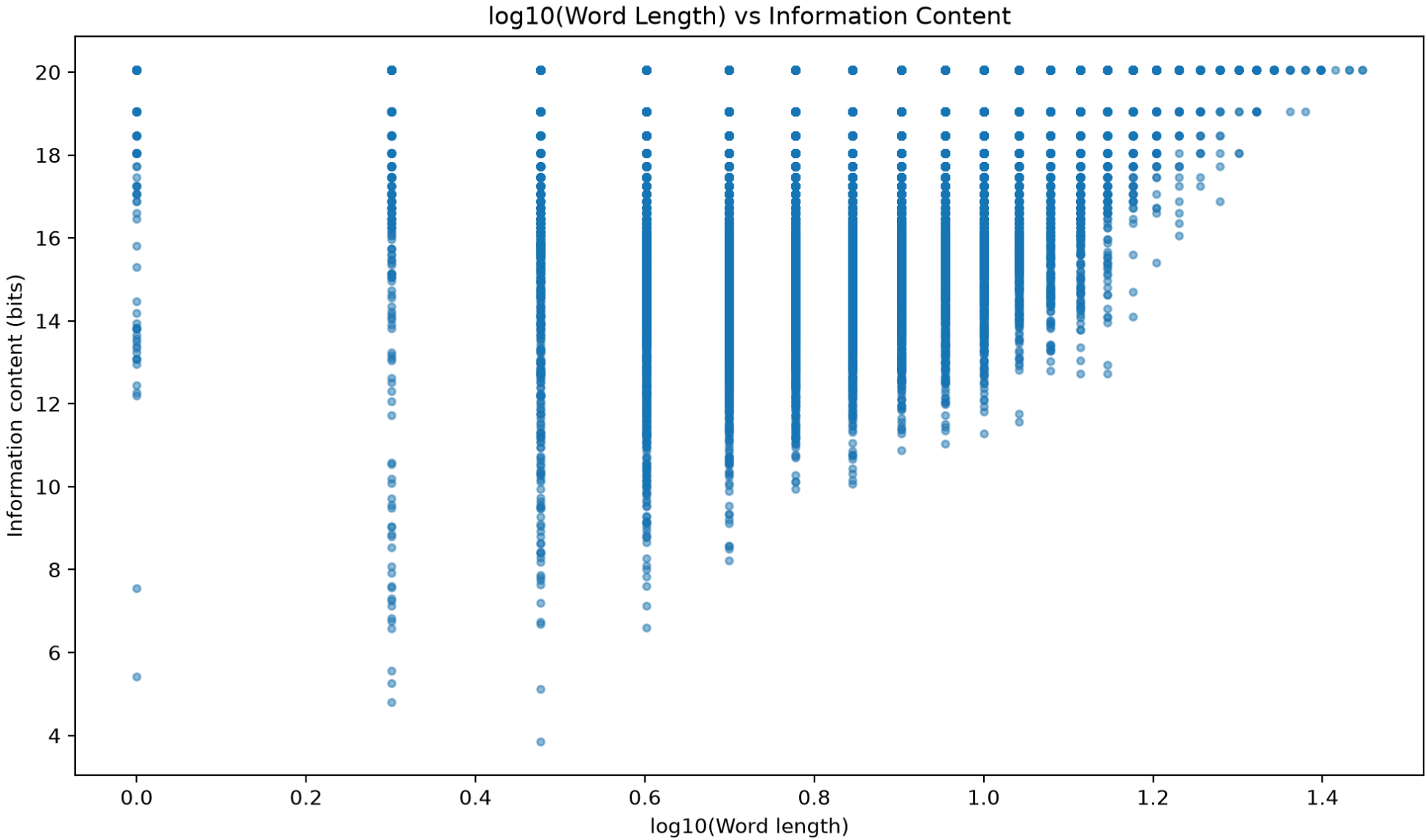
4) The graph of word length (x-axis) and frequency (y-axis) are as follows:



5) The graph of log10(word length) (x-axis) and log10(frequency) are as follows:



6) The graph of log10(word length) (x-axis) and information content (y-axis) is as follows:


log10(Word Length) vs Information Content
Information content (bits)
log10(Word length)
20
18
16
14
12
10
8
6
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

Q7)

Our results offer clear, if modest, evidence in favor of this idea — but the story is more nuanced than a simple “yes." (Small data size can be the reason like for the Q1 and 2).

Lets start with the intuitive version of the hypothesis: if language were shaped for efficient communication, the words we use constantly should be the cheapest to produce, and the words that carry rare, surprising information should be allowed to be longer.
Frequency is the obvious first place to look for this pattern, and it does show up — but barely. The Pearson correlation between word length and raw frequency was only $r = -0.053$. That's a real, statistically significant negative relationship (longer words are somewhat less frequent), but it's weak enough that frequency alone is clearly not doing much of the explanatory work. The regression confirms this numerically: frequency predicts almost nothing about length, with $R^2 = 0.00284$ — under 0.3% of the variance in word length.

Information content tells a different story. Once we reframe the predictor not as "how often is this word used" but as "how much information does this word carry" ($I(w) = -\log_2 P(w)$), the relationship strengthens considerably. The information-content model reached $R^2 = 0.05253$ — nearly 19 times the explanatory power of the frequency model. Visually, this shows up as a clear positive trend in the log(length) vs. information-content plot: words that are rarer and more surprising tend to run longer, while common, predictable words cluster at shorter lengths.
In other words, length isn't tracking usage frequency directly, but tracking how informative a word is, and frequency was only ever a rough tool kit for that.

This is precisely the reframing Piantadosi, Tily, and Gibson (2011) argue for and is worth mentioning. Their central claim is that word length is optimized not for frequency but for the amount of information a word conveys, and that frequency looks predictive of length mainly because it's correlated with information content, but not because frequency itself is what matters.
Using richer, context-sensitive estimates of information content (bigram and trigram surprisal from the BNC), they found correlations of 0.121 for frequency vs. length, rising to 0.161 with bigram information content and 0.168 with trigram information content — each step of context improving the fit. Our own jump from $R^2 = 0.003$ (frequency) to $R^2 = 0.053$ (unigram information content) is the same underlying effect, just measured with a much cruder tool.

That gap matters, because it points directly at what our analysis is missing. We estimated information content using only unigram (context-free) probabilities — essentially, "how rare is this word on average across the whole corpus." Piantadosi et al. instead estimated contextual predictability: how surprising a word is given the words immediately preceding it, using bigram and trigram models. That distinction is not a minor technical detail — it's the heart of their argument. A word like "bank" might be common overall (low unigram surprisal) but highly unpredictable in a specific sentence context (high contextual surprisal), and it's the contextual version that should govern how much "communicative work" that occurrence of the word is doing. Since our measure collapses this context-dependence away, we'd expect our R^2 to systematically underestimate the true strength of the information-content effect — which is consistent with our R^2 of 0.053 being well below what richer N-gram-based estimates typically recover.

The deeper motivation Piantadosi et al. give for why this pattern should exist is the idea of uniform information density: efficient communication isn't just about minimizing total effort, it's about avoiding sharp spikes and troughs in the rate at which information is transmitted. If a language allowed short forms for unpredictable, information-dense words, listeners would be bombarded with information faster than they could reliably process it at exactly the moments it mattered most. Keeping predictable words short and reserving longer forms for less predictable, more informative words smooths this out, spreading communicative load more evenly across an utterance rather than concentrating it in short bursts.
Taken together, our results are small-scale but directionally consistent with this account: information content clearly outperforms frequency as a predictor of word length in our data, even using the weakest possible (context-free) version of information content. That the effect strengthens further in Piantadosi et al.'s work once context is incorporated suggests our $R^2 = 0.053$ is likely a floor rather than a ceiling on how strongly information content actually governs word length — and reinforces their broader claim that human languages are shaped, at least in part, by pressure toward efficient, uniformly-paced communication rather than by raw usage frequency alone.

Final statement: The data leans “yes", word length reflects efficient-communication pressures, and specifically tracks information content over frequency — but the effect size here is modest, and richer context-aware measures would likely show it more strongly. (Some data shows a small divergence but is insignificant, while others just show a small alignment, not expected from a hypothesis that is strongly proven).

Github repo containing all the codes for Q3:
https://github.com/Bhargav170/2025114003_CL-2_Assignment-1